

M-Pesa Activity Tracker

"GitHub for your money activity"

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1. Overview

The M-Pesa Activity Tracker is a Flutter Android application that automatically parses M-Pesa SMS notifications, extracts structured transaction data, and visualises daily financial activity as a **GitHub-style contribution heatmap**.

Instead of scrolling through hundreds of SMS notifications or exporting statements manually, the app turns your raw M-Pesa history into a visual behavioural pattern — making spending habits and money movement immediately obvious at a glance.

"It's a personal financial contribution graph — but instead of code commits, it visualises your money activity from M-Pesa as daily behavioural patterns."

2. Features

2.1 Data Ingestion

Method	Description
SMS Sync	Reads M-Pesa messages directly from the device inbox with one tap
Paste SMS	Copy-paste one or many M-Pesa notifications into a text box
CSV Import	Import the official statement export from the M-Pesa app

2.2 Transaction Parsing

The parser recognises all standard M-Pesa transaction types:

- **Send Money** — transfers to phone numbers

- **Receive Money** — incoming transfers
- **Paybill** — utility and business payments
- **Buy Goods (Lipa na M-Pesa)** — till/merchant payments
- **Withdraw Cash** — agent withdrawals
- **Airtime Purchase** — top-up and data bundles
- **Reversal** — failed/reversed transactions

2.3 Heatmap Visualisation

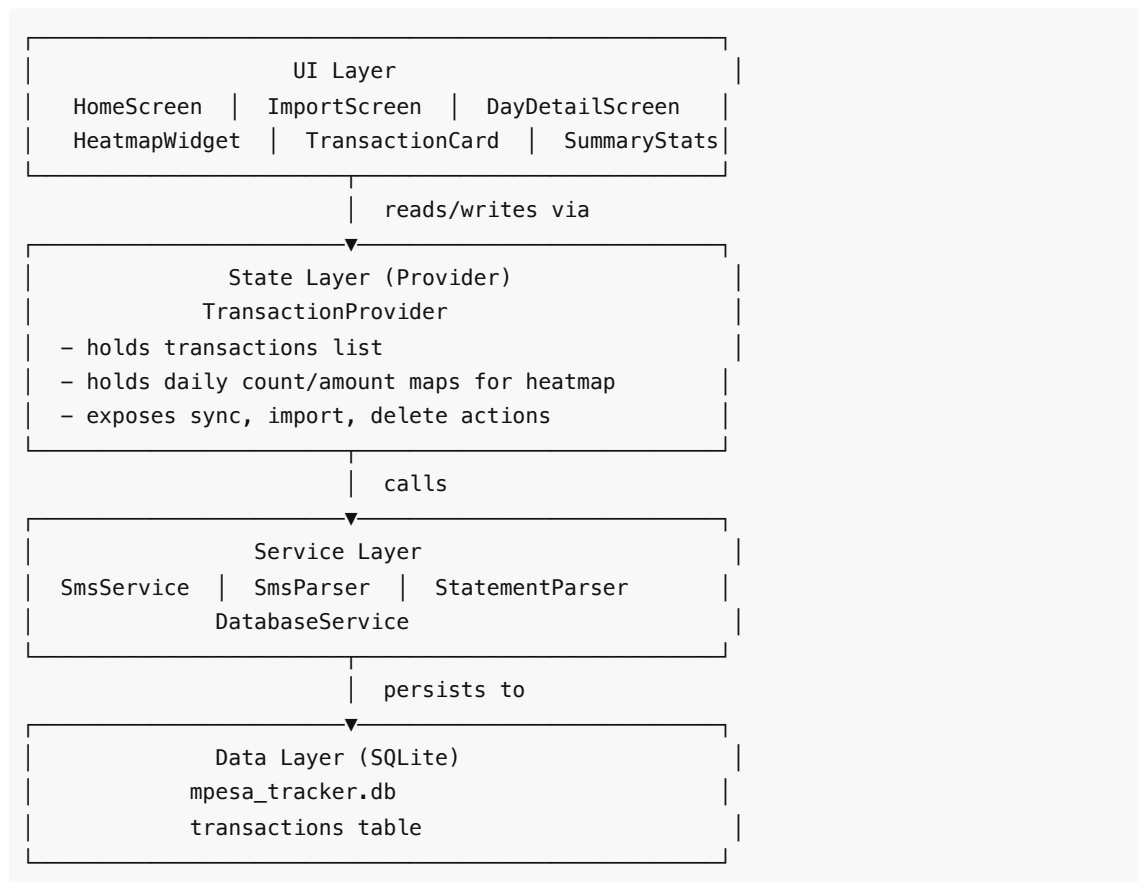
- GitHub-style 53-week grid (Monday → Sunday rows, oldest → newest columns)
- **5-level Safaricom-green intensity palette** based on daily transaction count
- Tap any cell to see every transaction on that day
- Year selector to navigate historical data
- Today's date highlighted with a white border

2.4 Transaction Management

- Full transaction list with swipe-to-delete
- Per-day detail view with net flow summary (received vs spent)
- Automatic deduplication — re-importing the same data never creates duplicates
- Clear all data option

3. Architecture

The app uses a clean layered architecture:



State Management: Provider (`ChangeNotifier`) — chosen for simplicity and testability without the overhead of BLoC or Riverpod for this app size.

Local Storage: SQLite via `sqflite` — durable, queryable, zero-config.

4. Project Structure

```
lib/
├─ main.dart                # Entry point, app theme, Provider setup
├─ models/
│   └─ transaction.dart    # MpesaTransaction, DaySummary, TransactionType
├─ services/
│   ├─ database_service.dart # SQLite singleton, CRUD, aggregation queries
│   ├─ sms_parser.dart      # Regex parser for M-Pesa SMS format
│   ├─ sms_service.dart     # Reads device inbox, permissions
│   └─ statement_parser.dart # CSV + pasted-SMS bulk import
├─ providers/
│   └─ transaction_provider.dart # ChangeNotifier, app-wide state
├─ screens/
│   ├─ home_screen.dart     # Main screen: heatmap + stats + list
│   ├─ import_screen.dart   # Tabbed import: paste SMS / CSV file
│   └─ day_detail_screen.dart # Per-day transaction breakdown
└─ widgets/
    ├─ heatmap_widget.dart  # Custom GitHub-style heatmap grid
    ├─ transaction_card.dart # Dismissible transaction list item
    └─ summary_stats.dart   # 3-tile stats row (count/spent/received)

android/
└─ app/src/main/
    └─ AndroidManifest.xml  # SMS + storage permissions

test/
└─ widget_test.dart
```

5. Data Flow

SMS Sync Flow

```
User taps Sync
  |
  ▼
SmsService.requestPermissions()
  |
  ▼ (granted)
flutter_sms_inbox reads device inbox
  |
  ▼
Filter: SmsParser.isMpesaSms(body)
  |
  ▼
```

```

SmsParser.parseAll(messages)
  - extract ref, amount, date, type, counterparty, balance
    |
    ▼
DatabaseService.insertTransactionsBatch()
  - UNIQUE(ref) deduplication
    |
    ▼
TransactionProvider refreshes heatmap data
    |
    ▼
UI rebuilds: HeatmapWidget + TransactionList

```

CSV Import Flow

```

User picks .csv file
  |
  ▼
dart:io File.readAsString()
  |
  ▼
StatementParser.parseCSV()
  - find header row (skip Safaricom metadata)
  - parse each row: receipt, date, paid in / withdrawn
  - infer transaction type from Details column
    |
    ▼
DatabaseService.insertTransactionsBatch()
  |
  ▼
UI refreshes

```

6. Key Components

6.1 TransactionProvider (providers/transaction_provider.dart)

The single source of truth for all UI state.

Property	Type	Description
transactions	List<MpesaTransaction>	All transactions, newest first
dailyCounts	Map<String, int>	"yyyy-MM-dd" → count, for heatmap
dailyAmounts	Map<String, double>	"yyyy-MM-dd" → total amount
selectedYear	int	Year displayed on heatmap
selectedDay	DateTime?	Day tapped on heatmap
state	LoadState	idle / loading / error

Key actions: `loadData()`, `syncSms()`, `importCsv()`, `importPastedSms()`, `deleteTransaction()`, `clearAll()`, `setSelectedYear()`, `selectDay()`

6.2 DatabaseService (`services/database_service.dart`)

Singleton SQLite wrapper. All queries are async-safe via sqflite's internal locking.

Notable methods:

```
// Batch insert with automatic dedup
Future<int> insertTransactionsBatch(List<MpesaTransaction> txs)

// Returns {yyyy-MM-dd: count} for an entire year – used to build heatmap
Future<Map<String, int>> fetchDailyCountsForYear(int year)

// Returns {yyyy-MM-dd: total_amount} for an entire year
Future<Map<String, double>> fetchDailyAmountsForYear(int year)
```

6.3 SmsParser (`services/sms_parser.dart`)

Pure Dart regex parser with no platform dependencies — fully unit-testable.

Extracts from each SMS message:

- **Reference** — the 10–12 character transaction code (e.g. `RHL92ABC12`)
- **Amount** — `Ksh1,234.00` → `1234.00`
- **Date & Time** — `1/5/26 at 2:30 PM` → `DateTime(2026, 5, 1, 14, 30)`
- **Type** — inferred from verb: *sent* / *received* / *withdrawn* / *paid* / *airtime*
- **Counterparty** — name and phone number after the verb
- **Balance** — new M-Pesa balance after transaction
- **Transaction cost** — fee charged

7. M-Pesa SMS Parsing

7.1 Supported SMS Formats

Send Money

```
RHL92ABC12 Confirmed. Ksh500.00 sent to JANE DOE 0722000000
on 1/5/26 at 2:30 PM. New M-Pesa balance is Ksh4,500.00.
Transaction cost, Ksh8.00.
```

Receive Money

```
RHL92ABC13 Confirmed.You have received Ksh1,000.00 from
JOHN KAMAU 0733000000 on 1/5/26 at 3:00 PM.
New M-Pesa balance is Ksh5,500.00.
```

Paybill Payment

```
RHL92ABC14 Confirmed. Ksh2,500.00 sent to SAFARICOM POSTPAY
100200 on 1/5/26 at 10:00 AM. New M-Pesa balance is Ksh3,000.00.
```

Transaction cost, Ksh0.00.

Buy Goods (Lipa na M-Pesa)

RHL92ABC15 Confirmed. Ksh350.00 paid to NAIVAS WESTGATE 12345 on 1/5/26 at 1:00 PM. New M-Pesa balance is Ksh2,650.00.

Withdraw Cash

RHL92ABC16 Confirmed. Ksh1,000.00 withdrawn from JOHN AGENT 000001 on 1/5/26 at 11:00 AM. New M-Pesa balance is Ksh1,650.00. Transaction cost, Ksh28.00.

Airtime Purchase

RHL92ABC17 Confirmed. Airtime purchase of Ksh100.00 on 1/5/26 at 12:00 PM. New M-Pesa balance is Ksh1,550.00.

7.2 Type Detection Logic

SMS body (lowercased)

contains "you have received" or "received ksh"	→ Received
contains "airtime purchase" or "airtime of"	→ Airtime
contains "withdrawn from"	→ Withdrawn
contains "reversal"	→ Reversal
contains "paid to"	→ Buy Goods
contains "sent to" + 5–6 digit account code	→ Paybill
contains "sent to" (phone number)	→ Sent

8. Database Design

Schema

```
CREATE TABLE transactions (  
  id          INTEGER PRIMARY KEY AUTOINCREMENT,  
  ref         TEXT      NOT NULL UNIQUE,    -- M-Pesa transaction ref  
  date        TEXT      NOT NULL,          -- ISO 8601 datetime  
  amount      REAL      NOT NULL,  
  type        TEXT      NOT NULL,          -- enum name  
  counterparty TEXT,                    -- name/number of other party  
  balance     TEXT,                    -- M-Pesa balance after tx  
  transaction_cost REAL,                -- fee charged  
  raw_sms     TEXT      NOT NULL          -- original SMS or import note  
);  
  
CREATE INDEX idx_date ON transactions(date);
```

Deduplication

Every M-Pesa transaction has a unique reference number (e.g. RHL92ABC12). The `UNIQUE` constraint on `ref` combined with `ConflictAlgorithm.ignore` means:

- Re-syncing SMS never creates duplicate records
- Re-importing the same CSV is safe
- Mixing sources (SMS + CSV of the same period) is safe

9. Heatmap Visualisation

Grid Layout

The heatmap renders a **53-column × 7-row** grid:

	Jan	Feb	Mar	...	Dec
Mon	[][][][][][][][][][][]...	[][][][][][][][][][][]			
Tue	[][][][][][][][][][][]...	[][][][][][][][][][][]			
Wed	[][][][][][][][][][][]...	[][][][][][][][][][][]			
Thu	[][][][][][][][][][][]...	[][][][][][][][][][][]			
Fri	[][][][][][][][][][][]...	[][][][][][][][][][][]			
Sat	[][][][][][][][][][][]...	[][][][][][][][][][][]			
Sun	[][][][][][][][][][][]...	[][][][][][][][][][][]			

- Each square = 1 day
- Columns = weeks (left = older, right = newer)
- Rows = Mon through Sun
- First/last weeks padded with empty cells to align

Colour Intensity

Level	Transactions/day	Colour
0	0	#1C1C1E (dark surface)
1	1–2	#0A3D1F (very dark green)
2	3–5	#1A6B38
3	6–10	#25A244
4	11+	#4CD964 (brightest)

The palette mirrors **Safaricom's brand green** at increasing brightness levels.

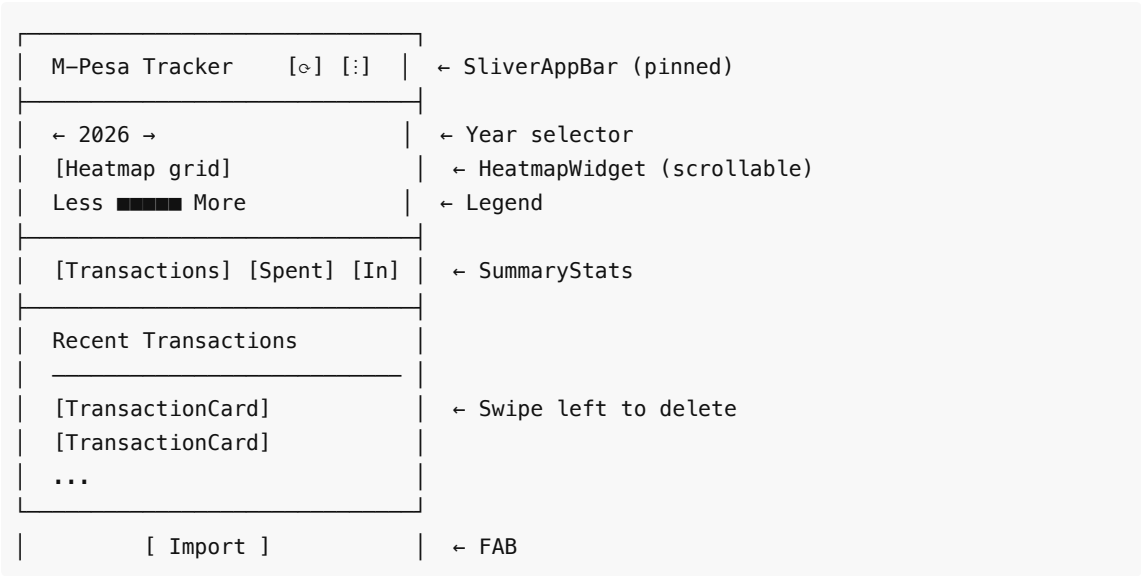
Interaction

- **Tap a cell** → navigates to `DayDetailScreen` for that date (if count > 0)
- **Today** → highlighted with a white border
- **Tooltip** → shows date and transaction count on long-press
- **Year selector** → `<` and `>` buttons rebuild the heatmap for any year

10. Screens & Navigation

HomeScreen

The main screen uses a CustomScrollView with SliverAppBar :



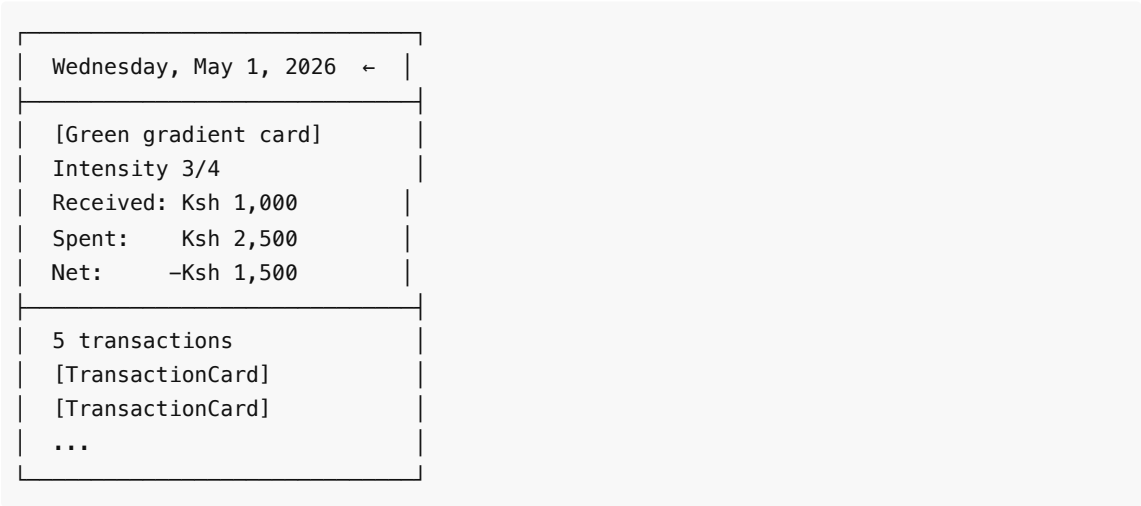
ImportScreen

Tabbed screen with two import methods:

- **Paste SMS tab** — multi-line text field, accepts one or many messages
- **CSV File tab** — file picker for .csv or .txt exports

DayDetailScreen

Drilled-in view for a single day:



11. Permissions

The app requests the following Android permissions at runtime:

Permission	Purpose	When requested
READ_SMS	Read M-Pesa messages from device inbox	On "Sync SMS" tap

RECEIVE_SMS	Declared for future real-time listening	On "Sync SMS" tap
READ_EXTERNAL_STORAGE	File import (Android ≤12)	On "Choose File" tap
READ_MEDIA_*	File import (Android 13+)	On "Choose File" tap

Import via Paste or CSV works completely **without any permissions** — the user manually copies the data.

12. Dependencies

Package	Version	Purpose
provider	^6.1.2	State management
sqflite	^2.3.3+1	Local SQLite database
path	^1.9.0	File path utilities
path_provider	^2.1.3	Platform-specific directories
file_picker	^8.1.2	Pick CSV/TXT files
permission_handler	^11.3.1	Runtime permissions
flutter_sms_inbox	^1.0.3	Read device SMS inbox
csv	^6.0.0	Parse CSV statement exports
intl	^0.19.0	Date & number formatting

13. Getting Started

Prerequisites

- Flutter 3.41+ (flutter --version)
- Android device or emulator (API 21+)
- Android Studio or VS Code with Flutter extension

Build & Run

```
# Clone the repository
git clone https://github.com/lawmaluki/M-Pesa-Activity-Tracker.git
cd M-Pesa-Activity-Tracker

# Install dependencies
flutter pub get

# Run on connected Android device
flutter run
```

```
# Build release APK
flutter build apk --release
```

First Use

1. **Launch the app** — the heatmap starts empty
2. **Import data** using any of three methods:
 - Tap **Sync SMS** (top-right ☰) → grant SMS permission → syncs all M-Pesa inbox messages automatically
 - Tap **Import FAB** → **Paste SMS** tab → paste copied M-Pesa messages → tap Import
 - Tap **Import FAB** → **CSV File** tab → choose your Safaricom statement export
3. **Tap any green cell** on the heatmap to see that day's transactions
4. **Use the year selector** (< 2026 >) to browse past years

14. Future Extensions

Feature	Description
Real-time SMS listener	Native Android BroadcastReceiver via platform channel — auto-adds new M-Pesa messages as they arrive
Monthly activity score	Roll up spending/receiving into a single behavioural score per month
Category breakdown	Pie/bar chart of spend by type (Paybill, Buy Goods, Airtime, etc.)
Budget alerts	Set daily/weekly spending limits with notifications
Export to CSV	Re-export your parsed transaction history
Multi-account	Track multiple Safaricom lines
Full finance dashboard	Balance trends, net worth tracking, savings goals

M-Pesa Activity Tracker — built with Flutter. For personal use.